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$$f(x) = \sin(2x)$$

$$f'(x) = 2 \cos(2x)$$

$$f'(x) = 2 \sin\left(2x + \frac{\pi}{2}\right)$$

$$f''(x) = 2^2 \cos\left(2x + \frac{\pi}{2}\right)$$

$$f''(x) = 2^2 \sin(2x + \pi)$$

$$f'''(x) = 2^3 \cos(2x + \pi)$$

$$f'''(x) = 2^3 \sin\left(2x + \frac{3\pi}{2}\right)$$

$\vdots$

$$f^{(n)}(x) = 2^n \sin\left(2x + \frac{n\pi}{2}\right)$$

References